where $\widetilde{\boldsymbol{u}}_{j,c_j} := \boldsymbol{u}_{j,c_j} - \bar{\boldsymbol{u}}_j$ is the centered factor for concept j for the value c_j, and $\bar{\boldsymbol{u}}_j := \frac{1}{2}(\boldsymbol{u}_{j,1} + \boldsymbol{u}_{j,0})$ is the average of the two concept factors.

But the centered factors must sum to zero, so $\widetilde{\boldsymbol{u}}_{j,0} = -\widetilde{\boldsymbol{u}}_{j,1}$. Therefore for any $i \neq j$

$$\begin{aligned} & (\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0})^\intercal (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0}) = 0 \\ \Rightarrow \quad & (\widetilde{\boldsymbol{u}}_{i,1} - \widetilde{\boldsymbol{u}}_{i,0})^\intercal (\widetilde{\boldsymbol{u}}_{j,1} - \widetilde{\boldsymbol{u}}_{j,0}) = 0 \\ \Rightarrow \quad & \widetilde{\boldsymbol{u}}_{i,1}^\intercal \widetilde{\boldsymbol{u}}_{j,1} = 0. \end{aligned} \tag{58}$$

By evaluating the classification rule at any $\boldsymbol{z_c}$ for any concept i we get

$$\boldsymbol{w}_i^\intercal \boldsymbol{z_c} + b_i = \boldsymbol{w}_i^\intercal \left(\sum_{j=1}^{k} \left(\widetilde{\boldsymbol{u}}_{j,c_j} + \bar{\boldsymbol{u}}_j \right) \right) + b_i \tag{59}$$

$$= \boldsymbol{w}_i^\intercal \sum_{j=1}^{k} \widetilde{\boldsymbol{u}}_{j,c_j} + \boldsymbol{w}_i^\intercal \sum_{j=1}^{k} \bar{\boldsymbol{u}}_j + b_i, \tag{60}$$

but note that only the first term of (60) is affected by the concept i, and the following terms can be absorbed into the bias b_i[2]. Thus, the classification is correct only if $\text{sign}\left(\boldsymbol{w}_i^\intercal \sum_{j=1}^{k} \widetilde{\boldsymbol{u}}_{j,c_j}\right) = 2c_i - 1$. By (56) it follows that

$$\text{sign}\left(\boldsymbol{w}_i^\intercal \sum_{j=1}^{k} \widetilde{\boldsymbol{u}}_{j,c_j} \right) = \text{sign}\left((2\widetilde{\boldsymbol{u}}_{i,1})^\intercal \widetilde{\boldsymbol{u}}_{i,c_i} \right) = \text{sign}\left(\widetilde{\boldsymbol{u}}_{i,1}^\intercal \widetilde{\boldsymbol{u}}_{i,c_i} \right) = 2c_i - 1. \tag{61}$$

And therefore the classification is correct.

(1) $|T| = 2^{k-1} + 1$ case. The main idea of the proof is that for any concept $i \in [k]$ and a pair of "counterfactual" points $\boldsymbol{z_c}, \boldsymbol{z}_{\boldsymbol{c}(i \to \bar{c}_i)}$ it holds that the distance between the two points is the shortest possible distance between any two points in the convex hulls of the two classes and is equal to $\|\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}\|^2$. This follows the same argument as specified in the full dataset case (and in (56) in particular). All that is needed to be shown then, is the availability of such a pair of points in any training set with $|T| = 2^{k-1} + 1$.

This can be argued by contradiction. Assume that for some concept $i \in [k]$ there are no two points $(\boldsymbol{c}, \boldsymbol{c}')$ with $c_j = c'_j$ for all $j \neq i$ and $c_i \neq c'_i$. There are 2^{k-1} such pairs of points in total. Thus a dataset would have to have at most $2^k - 2^{k-1} = 2^{k-1}$ points, which contradicts the assumption that $|T| = 2^{k-1} + 1$. Therefore, such a pair of points must exist. Since this argument is independent of the particular choice of the concept i, it holds for any concept. Therefore the SVM solution will always be able to find such a pair of points that minimize the distance between the two classes (as per (54)) and will transfer as well as yield the same weight vector as the full dataset case.

(2) $\mathcal{D}^c$ case with $|T| = 1 + k$. By construction for any concept i there exists "counterfactual" point from the center $\boldsymbol{c}$ to $\boldsymbol{c}(i \to \bar{c}_i)$. By following the same argument detailed in the full dataset case as well as the previous case, it follows that the SVM solution will transfer as well as yield the same weight vector as the full dataset case. In both cases stability of the solution follows due to recovering the same weight and bias vectors. □

D.2. General case: linearly factored embeddings and sufficiency of recovering the factors

We provide a complementary analysis on the sufficient conditions for generalizing compositionally. Here, we detail the key results for recovering the factors $\boldsymbol{u}$ from representations that already possess linear factorization.

We first note the minimal dataset setting using the notion of a cross dataset, defined below.

Lemma 5 (Uniqueness up to concept-wise shifts)**.** Let the concept space be $\mathcal{C} = \mathcal{C}_1 \times \cdots \times \mathcal{C}_k$. Assume two factor families $\{\boldsymbol{u}_{i,j}\}_{i \in [k],\, j \in \mathcal{C}_i}$ and $\{\boldsymbol{v}_{i,j}\}_{i \in [k],\, j \in \mathcal{C}_i}$ satisfy, for every $\boldsymbol{c} \in \mathcal{C}$,

$$\boldsymbol{z_c} = \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i} = \sum_{i=1}^{k} \boldsymbol{v}_{i,c_i}.$$

Then there exist vectors $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k \in \mathbb{R}^d$ with

$$\sum_{i=1}^{k} \boldsymbol{s}_i = \boldsymbol{0}$$

such that

$$\boldsymbol{v}_{i,j} = \boldsymbol{u}_{i,j} + \boldsymbol{s}_i, \qquad \forall i \in [k],\ j \in \mathcal{C}_i.$$

Hence the factors are identifiable only up to concept-wise shifts.

[2] Alternatively, assume that $\boldsymbol{z_c}$ is zero-centered and use the same argument.

Proof. For each concept $i \in [k]$ and value $j \in \mathcal{C}_i$, define the vector difference

$$\boldsymbol{\delta}_{i,j} := \boldsymbol{v}_{i,j} - \boldsymbol{u}_{i,j}.$$

We first show that, within each concept i, this difference is independent of the value. Fix $i \in [k]$ and pick any two values $p, q \in \mathcal{C}_i$. Take any $\boldsymbol{c} \in \mathcal{C}$ with $c_i = p$. Using the counterfactual tuple $\boldsymbol{c}(i \to q)$ and subtracting the two factorizations gives

$$\begin{aligned} \boldsymbol{z}_{\boldsymbol{c}} - \boldsymbol{z}_{\boldsymbol{c}(i\to q)} &= \sum_{\ell=1}^{k} \boldsymbol{u}_{\ell,c_\ell} - \sum_{\ell=1}^{k} \boldsymbol{u}_{\ell,(\boldsymbol{c}(i\to q))_\ell} \\ &= \sum_{\ell=1}^{k} \boldsymbol{v}_{\ell,c_\ell} - \sum_{\ell=1}^{k} \boldsymbol{v}_{\ell,(\boldsymbol{c}(i\to q))_\ell}. \end{aligned}$$

All terms except concept i cancel on both sides, so

$$\boldsymbol{u}_{i,p} - \boldsymbol{u}_{i,q} = \boldsymbol{v}_{i,p} - \boldsymbol{v}_{i,q}, \qquad \Rightarrow \qquad \boldsymbol{\delta}_{i,p} = \boldsymbol{\delta}_{i,q}.$$

Hence, for each concept i, there is a single vector $\boldsymbol{s}_i \in \mathbb{R}^d$ such that

$$\boldsymbol{\delta}_{i,j} = \boldsymbol{s}_i, \qquad \forall j \in \mathcal{C}_i.$$

Therefore $\boldsymbol{v}_{i,j} = \boldsymbol{u}_{i,j} + \boldsymbol{s}_i$ for all i, j.

To obtain the zero-sum constraint, evaluate at any $\boldsymbol{c} \in \mathcal{C}$:

$$\mathbf{0} = \sum_{i=1}^{k} \left(\boldsymbol{v}_{i,c_i} - \boldsymbol{u}_{i,c_i}\right) = \sum_{i=1}^{k} \boldsymbol{\delta}_{i,c_i} = \sum_{i=1}^{k} \boldsymbol{s}_i.$$

Conversely, if $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k \in \mathbb{R}^d$ satisfy $\sum_{i=1}^{k} \boldsymbol{s}_i = \mathbf{0}$ and we set $\boldsymbol{v}_{i,j} := \boldsymbol{u}_{i,j} + \boldsymbol{s}_i$, then for every $\boldsymbol{c} \in \mathcal{C}$,

$$\sum_{i=1}^{k} \boldsymbol{v}_{i,c_i} = \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i} + \sum_{i=1}^{k} \boldsymbol{s}_i = \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i},$$

so the reconstructed factors generate the same embeddings. □

We illustrate this lemma graphically in Figure 14.

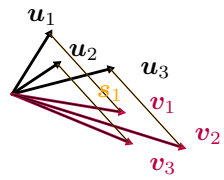

Figure 14. **Illustration of the shift ambiguity in the factorization equations.** The black factors $\{\boldsymbol{u}_i\}$ and pink factors $\{\boldsymbol{v}_i\}$ produce the same pairwise differences; the orange arrows show concept-wise shift vectors $\boldsymbol{s}_i$ such that $\boldsymbol{v}_i = \boldsymbol{u}_i + \boldsymbol{s}_i$.

A neat consequence is that centered factors are uniquely determined: any recovered factorization, once centered per concept, matches the true centered factors.

Corollary 1 (Uniqueness of centered factors)**.** Assume the setting of Lemma 5. For each concept i, let

$$\boldsymbol{\mu}_i := \frac{1}{|\mathcal{C}_i|} \sum_{j \in \mathcal{C}_i} \boldsymbol{u}_{i,j},$$

$$\boldsymbol{\nu}_i := \frac{1}{|\mathcal{C}_i|} \sum_{j \in \mathcal{C}_i} \boldsymbol{v}_{i,j},$$

and define the centered factors $\boldsymbol{u}^\circ_{i,j} := \boldsymbol{u}_{i,j} - \boldsymbol{\mu}_i$ and $\boldsymbol{v}^\circ_{i,j} := \boldsymbol{v}_{i,j} - \boldsymbol{\nu}_i$. Then $\boldsymbol{u}^\circ_{i,j} = \boldsymbol{v}^\circ_{i,j}$ for all $i \in [k]$ and $j \in \mathcal{C}_i$. Equivalently, for every $\boldsymbol{c} \in \mathcal{C}$,

$$\sum_{i=1}^{k} \boldsymbol{u}^\circ_{i,c_i} = \sum_{i=1}^{k} \boldsymbol{v}^\circ_{i,c_i},$$

so the centered factorization is unique.

Proof. By Lemma 5, there exist $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k$ with $\sum_i \boldsymbol{s}_i = \mathbf{0}$ such that $\boldsymbol{v}_{i,j} = \boldsymbol{u}_{i,j} + \boldsymbol{s}_i$ for all i, j. Averaging over $j \in \mathcal{C}_i$ yields $\boldsymbol{\nu}_i = \boldsymbol{\mu}_i + \boldsymbol{s}_i$. Thus,

$$\begin{aligned}
\boldsymbol{v}^{\circ}_{i,j} &= \boldsymbol{v}_{i,j} - \boldsymbol{\nu}_i \\
&= (\boldsymbol{u}_{i,j} + \boldsymbol{s}_i) - (\boldsymbol{\mu}_i + \boldsymbol{s}_i) \\
&= \boldsymbol{u}_{i,j} - \boldsymbol{\mu}_i \\
&= \boldsymbol{u}^{\circ}_{i,j},
\end{aligned}$$

as claimed. □

First, we consider the general case where concept directions are not necessarily linearly independent. Suppose the inputs $\boldsymbol{z_c}$ are linearly separable for any $i \in [k], j \in [n]$. If we can recover all factors $\boldsymbol{v}_{i,j}$, we can reconstruct $\boldsymbol{z_c} = \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i}$ and then fit linear probes for the concept values.

By Lemma 5 (and the centered-factor corollary above), recovery is unique up to concept-wise shifts, so the key requirement is rank of the one-hot design matrix induced by observed tuples.

Proposition 5 (Maximum possible rank of the design matrix)**.** Let $\boldsymbol{A} \in \{0,1\}^{n^k \times kn}$ be the design matrix whose kn columns are $\{\boldsymbol{a}_{i,r} : i = 1, \ldots, k,\ r = 1, \ldots, n\}$, arranged in k blocks of size n, with all n^k treatment combinations as rows and each row having exactly one 1 in each block. Then,

$$\operatorname{rank}(\boldsymbol{A}) = 1 + k(n-1).$$

Proof. We first show the upper bound by spanning columns. Define $\mathbf{1} \in \mathbb{R}^{n^k}$ as the all-ones vector and, for each block i and each $r = 2, \ldots, n$,

$$\boldsymbol{d}_{i,r} := \boldsymbol{a}_{i,r} - \boldsymbol{a}_{i,1}.$$

Let

$$\mathcal{B} := \{\mathbf{1}\} \cup \{\boldsymbol{d}_{i,r} : 1 \le i \le k,\ 2 \le r \le n\}, \quad \text{so} \quad |\mathcal{B}| = 1 + k(n-1).$$

For every block i, $\sum_{r=1}^{n} \boldsymbol{a}_{i,r} = \mathbf{1}$, hence

$$\sum_{r=2}^{n} \boldsymbol{d}_{i,r} = \mathbf{1} - n\boldsymbol{a}_{i,1} \ \Rightarrow\ \boldsymbol{a}_{i,1} = \tfrac{1}{n}\left(\mathbf{1} - \sum_{r=2}^{n} \boldsymbol{d}_{i,r}\right), \quad \boldsymbol{a}_{i,r} = \boldsymbol{a}_{i,1} + \boldsymbol{d}_{i,r}\ (r \ge 2).$$

Thus every original column $\boldsymbol{a}_{i,r}$ lies in $\operatorname{span} \mathcal{B}$, so $\operatorname{rank}(\boldsymbol{A}) \le 1 + k(n-1)$.

For the matching lower bound, note that $\boldsymbol{A}^{\intercal}$ is exactly the on-off matrix from Proposition 6 with $\alpha = 1, \beta = 0$ (rows indexed by (i, r) and columns by tuples $\boldsymbol{c}$). The rank argument in that proof applies verbatim and gives $\operatorname{rank}(\boldsymbol{A}^{\intercal}) = 1 + k(n-1)$. Hence

$$\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}^{\intercal}) = 1 + k(n-1).$$

□

When $\operatorname{rank}(\boldsymbol{A}) = 1 + k(n-1)$, the linear system $\boldsymbol{Z} = \boldsymbol{A}\boldsymbol{U}$ determines the centered factors uniquely (up to the shift ambiguity already characterized above). Therefore one can reconstruct the grid embeddings and, under linear separability, fit linear readouts that recover concept values on all combinations (Definition 3).